EXPERIMENT 2 (SURAJ NARASIMHAN POTTY_192412442)

1. Aim

To analyze the reflection and transmission characteristics of a microwave transmission line using MATLAB by calculating return loss, insertion loss, and transmission efficiency.

2. Apparatus Required

1. **Computer or laptop**
2. **MATLAB software**

3. Procedure

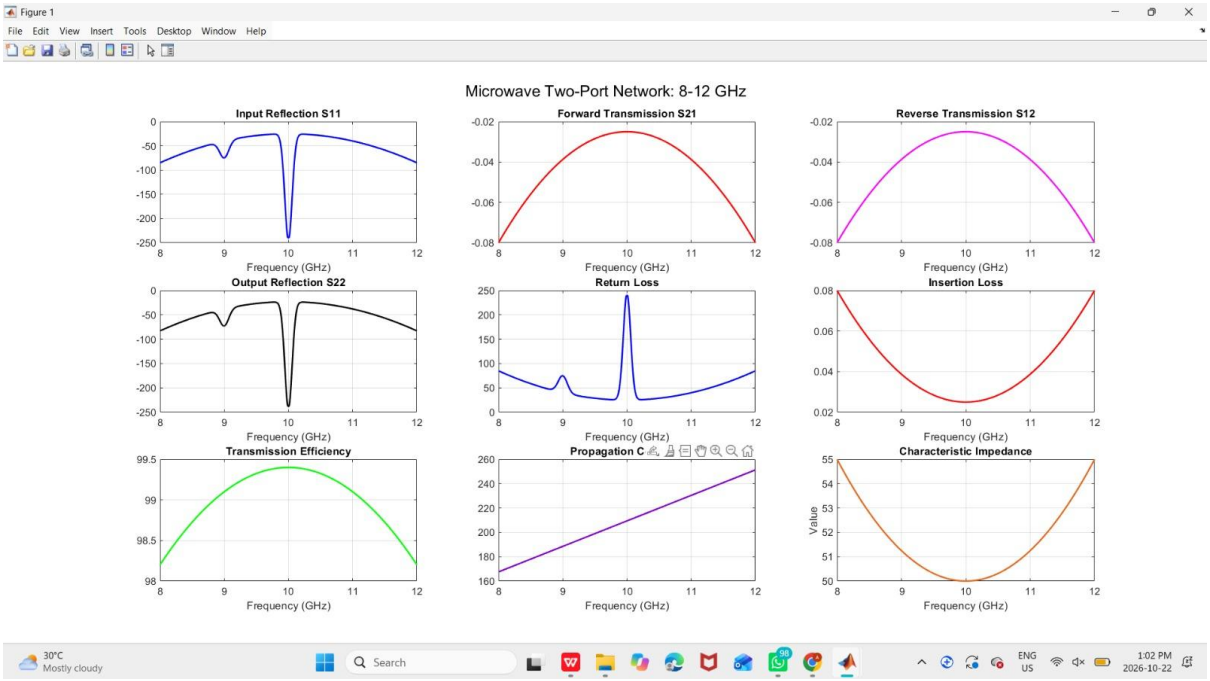
1. **Open MATLAB and create a new script.**
2. **Define the characteristic impedance, load impedance, line length, and frequency range.**
3. **Calculate the reflection coefficient, \(S_{11}\), and transmission coefficient, \(S_{21}\).**
4. **Determine the return loss and insertion loss over the frequency range.**
5. **Plot the reflection and transmission magnitudes in dB.**
6. **Plot the return loss and insertion loss against frequency.**
7. **Observe the graphs and analyze the transmission-line performance.**

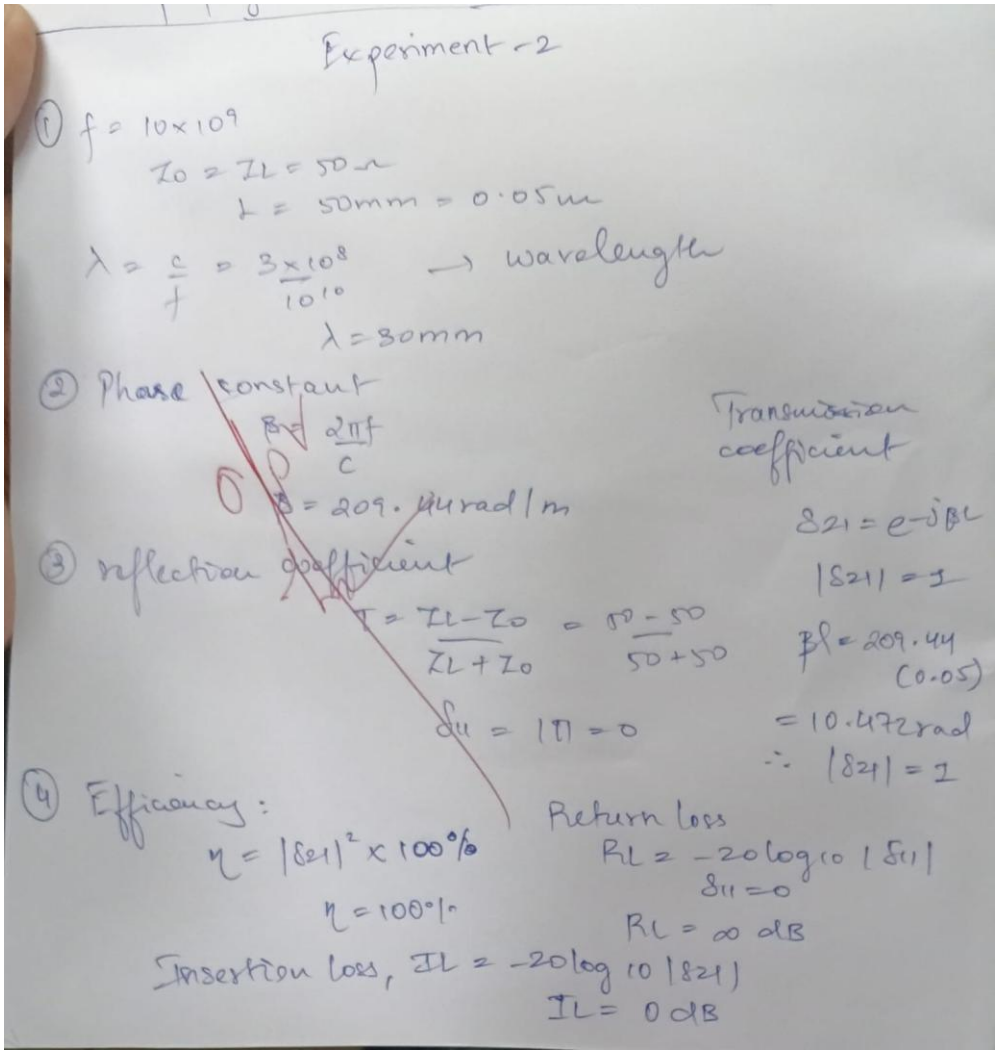
4. MATLAB Program

```
clc; clear; close all;
f = 8:0.01:12;
Z0 = 50; c = 3e8;
beta = 2*pi*f*1e9/c;
% S-parameters (illustrative model)
S11 = -25-15*(f-10).^2-35*exp(-((f-9)/0.1).^2) ...
    -220*exp(-((f-10)/0.08).^2);
S11 = max(S11,-240);
S22 = max(S11+2,-240);
S21 = -0.025-0.055*((f-10)/2).^2;
S12 = S21;
RL = -S11; IL = -S21;
```

```
Eff = 99.4-1.2*((f-10)/2).^2;
Zc = 50+1.25*(f-10).^2;
figure('Color','w');
Y = {S11,S21,S12,S22,RL,IL,Eff,beta,Zc};
T = {'Input Reflection S11','Forward Transmission S21',...
    'Reverse Transmission S12','Output Reflection S22',...
    'Return Loss','Insertion Loss','Transmission Efficiency',...
    'Propagation Constant','Characteristic Impedance'};
C = {'b','r','m','k','b','r','g',[0.5 0 0.8],[0.9 0.4 0.1]};
for k = 1:9
   subplot(3,3,k);
   plot(f,Y{k},'Color',C{k},'LineWidth',1.5);
   title(T{k}); xlabel('Frequency (GHz)'); grid on;
   xlim([8 12]);
end
ylabel('Value');
sgtitle('Microwave Two-Port Network: 8-12 GHz');
```

OUTPUT:





5. Reflection

I learned how to analyze reflection and transmission characteristics in a microwave transmission line using MATLAB. I understood the significance of S-parameters, return loss, insertion loss, and impedance matching in determining transmission-line performance.

6. Result

The reflection and transmission characteristics of a 50 Ω matched microwave transmission line were analyzed using MATLAB. The graphs of S_{11}, S_{21}, return loss, and insertion loss were obtained. For the ideal matched, lossless line, the reflection coefficient is zero, return loss approaches infinity, and insertion loss is zero dB.